# Symbiosis Skills and Professional University Kiwale, Pune

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**PROJECT REPORT**

**On**

**"FOREST FIRE PREDICTION"**

**Submitted by**

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# STUDENT DECLARATION AND ATTESTATION BY TRAINER

This is to declare that this report has been written by me. No part of the report is plagiarized from other sources. All information included from other sources have been duly acknowledged. I aver that if any part of the report is found to be plagiarized, I shall take full responsibility for it.

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# CERTIFICATE

This is to certify that the report entitled, "***Forest Fire Prediction***" submitted by"**Akshay Parulekar, Swapnil Patil, Yash Manchekar & Yukta Shinde**" to Symbiosis Skills and Professional University, Pune, Maharashtra, India, is a record of bonafide Project work carried out by him under my supervision and guidance and is worthy of consideration for the completion of certificate course in 'Machine Learning Engineer".

Signature of Trainer
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Date: / / 2021

_____ _____

Supervisor Supervisor

Date:

# ACKNOWLEDGEMENT

After the completion of this work, words are not enough to express feelings about all those who helped us to reach goal. It's a great pleasure and moment of immense satisfaction for us to express my profound gratitude to Project Guide, Mr. Kushal Sharma sir, whose constant encouragement enabled us to work enthusiastically. His perpetual motivation, patience and excellent expertise in discussion during progress of the project work have benefited us to an extent, which is beyond expression. We are happy to the part of School of Data Science, Symbiosis Skill Professional University and learning the new things with skilled professionals. We are thankful to Principal of Symbiosis Skills and Professional University Kiwale, Pune, India for providing an outstanding academic environment, also for providing the adequate facilities.

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# Forest Fire Prediction through EDA and Machine learning

## ABSTRACT

As we all know, the forest is considered as one of the most important and indispensable resources and Forest fires represent a constant threat to ecological systems, infrastructure and environmental aspects of a community, forest fire detection is a very important issue in the pre-suppression process. This gives rise to the urgent need to predict forest fires as soon as possible. This paper highlights the powerful feature of sensor networks as a potential solution to the challenge of early detection of forest fires. The algorithm presented makes use of data from various sensors attached, to fulfil the task in question. These collected data are transmitted to Hadoop HDFS where they are analysed. The proposed solution is based on sensor data and analysis, performs early prediction of any fire threat that might occurs in the forest.

# INTRODUCTION

Forest fires, also known as wild fires, are uncontrolled fires occurring in wild areas and cause significant damage to natural and human resources. It is an uncontrolled fire in an area of combustible vegetation that occurs in the countryside or in forest area. India witnessed the most severe forest fires in the recent time during the summer of 1995 in the hills of Uttar Pradesh and Himachal Pradesh. The Forest Survey of India's data on forest fire attributes around 50% of the forest areas as fire prone.

Forest fires often start unnoticed and spread very quickly, causing millions of dollars in damage and claiming many human lives every year in the many countries. Early prediction of hot spots and the initiation of appropriate measures can prevent, or, at least minimize damage and casualties. Common causes of forest fires are lightning, extreme hot and arid weather, severe drought, and human unawareness. Current satellite-imagery-based forest fire detection systems cannot detect forest fires with high precision and accuracy. A sensor-based forest fire detection system has the potential to achieve the high detection resolution and accuracy that is required for early detection of forest fires.

# LITERATURE SURVEY

Forest is considered as one of the most important and indispensable resource, furthermore, as the protector of the Earth's ecological balance. However, forest fire, affected by some human uncontrolled behaviour in social activities and abnormal natural factors, occurs occasionally. Forest fire was considered as one of the severest disasters.

In forest fire detection, it is essential to know how fire affects the soil mantle, stems and treetops, as well as how to detect underground fires. The sensor network must cover large areas, distributing high amount of sensing nodes, inexpensive sensors are needed to achieve cost reduction.

Detection capabilities is negatively influenced by the fact that often fires are not directly visible from the sensor because during the first phases they grow up in the underbrush and are occluded from the vegetation.

The most critical issue in a forest fire detection system is immediate response in order to minimize the scale of the disaster. This requires constant surveillance of the forest area. Current medium and large-scale fire surveillance systems do not accomplish timely detection due to low resolution and long period of scan. Therefore, there is a need for a scalable solution that can provide real-time fire detection with high accuracy. We believe that sensor networks can potentially provide such solution. Recent advances in sensor networks support our belief that they make a promising framework for building near real time forest fire detection systems. Currently, sensing modules can sense a variety of phenomena including temperature, relative humidity, oxygen, CO, CO2 and smoke which are all helpful for fire detection systems.

## Purpose of Project

To design a system to predict possible occurrences of forest fire using machine learning with help of data acquired from various wired/wireless sensors.

## Period of Project

12 weeks

# PROBLEM STATEMENT

The most common hazard in forests is forests fire. Forests fires are as old as the forests themselves. They pose a threat not only to the forest wealth but also to the entire regime to fauna and flora seriously disturbing the bio-diversity and the ecology and environment of a region. During summer, when there is no rain for months, the forests become littered with dry senescent leaves and twinges, which could burst into flames ignited by the slightest spark. The Himalayan forests, particularly, Garhwal Himalayas have been burning regularly during the last few summers, with colossal loss of vegetation cover of that region.

## Impacts of Forest Fires on Biological Environment

Forest fires also pose serious health hazards by producing smoke and noxious gases, similar to the events in Indonesia after the forest fires on the islands of Sumatra and Borneo in 1977 have shown. The burning of vegetation gives off not only carbon dioxide but also a host of other noxious gases (Greenhouse gases) such as carbon monoxide, methane, hydrocarbons, nitric oxide and nitrous oxide, that lead to global warming and ozone layer depletion. Burning forests and grasslands also add to already serious threat to the environment and also in global warming.

## Forest Fire Detection System

It is well known, there will be large variations/increase in temperature from the normal temperature whenever forest fire occurs. This can be detected/ monitored continuously by using temperature sensors and in accordance with the simple arrangement of transmitters. This concept is quite reliable and cost-effective in detection of forest fires since simple equipment’s are arranged in a simple configuration.

## System Description

The proposed system consists of stationary Wireless Sensor Networks (WSN) that monitor the designated forest area. It monitors the environmental conditions and assesses the threats on the basis of the parameters such as temperature, wind speed and wind direction. All data are fused together at an automated Data Lake ecosystem. If a threat is detected with a risk assessment rating above a predefined threshold, interested parties such as municipality, forest protection service and fire

brigade officers will get notified for every step of the procedure and give their consent for critical steps of the procedure. The system can be a powerful tool that enable the key authorities to take decisions based on accurate data.

## SYSTEM DESIGN

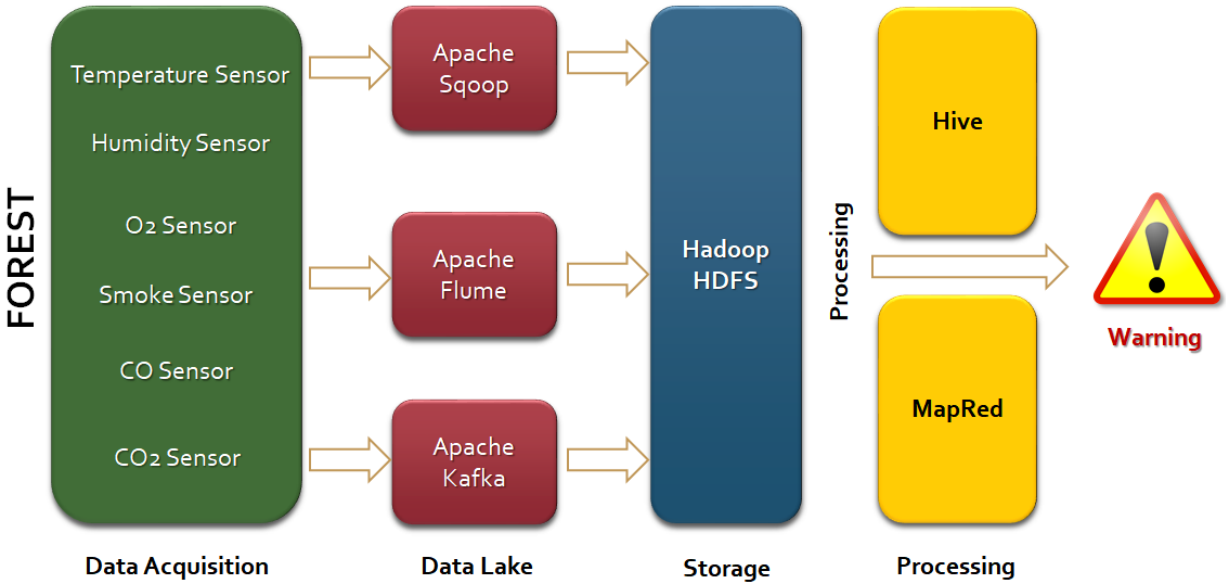
Mainly this project contains following subsystems:

1) Data Acquisition from each sensor
2) Data Lake
3) Database system
4) Storage (Hadoop HDFS)
5) Analysis Modul (MapReduce)

## IMPLEMENTATION

All Sensor Setup is kept in a box which is designed to withstand a high temperature and are located few feet above the ground. It consists of Wired/Wireless temperature sensor, humidity sensor, oxygen sensor, CO sensor, CO2 sensor and smoke sensor. This arrangement is connected to a data transmitter.
The function of the data transmitter is to transmit the data/signals from sensors to the data lake.
The data from the data lake will be migrated to Hadoop HDFS system after pre-processing.
HDFS receives all the data and fetch it to the analysis systems like Apache Hive or MapReduce.

## ARCHITECTUR



## WORKING

As shown in architecture diagram, we pulled data from various sensors.

***Data Lake-*** Sensors like temperature sensor, humidity sensor, anemometer and other auxiliary sensors gets information from the forest environment. Temperature sensor gave data about the warmness of the environment in the forest in terms of degree Celsius unit. Humidity sensor gave the relative humidity reading to the system. Anemometer gave the wind speed and also direction of the same. We collected that data from various sensors and made a data lake using some data migratory tools like Apache Sqoop, Apache Flume and Apache Kafka. Data from those sensors were migrated using Apache Sqoop.

***Data Storage-*** In this stage, collected data in Data Lake has been transferred to Hadoop Distributed File System which is Hadoop's storage part. We collected all the data in HDFS for further processing part.

***Processing-*** The final stage of our model in which we could have used Apache Hive OR MapReduce. We picked MapReduce for the final processing of our data. For coding, we have been using Python and for this project we used Pydoop as a major part of python because Pydoop combines Python and Hadoop. After successful processing of the data received it will give warning according to the circumstances.

## TOOLS

Python Programming

Hadoop | HDFS

Apache Sqoop | Apache Flume | Apache Kafka (Data Migration)

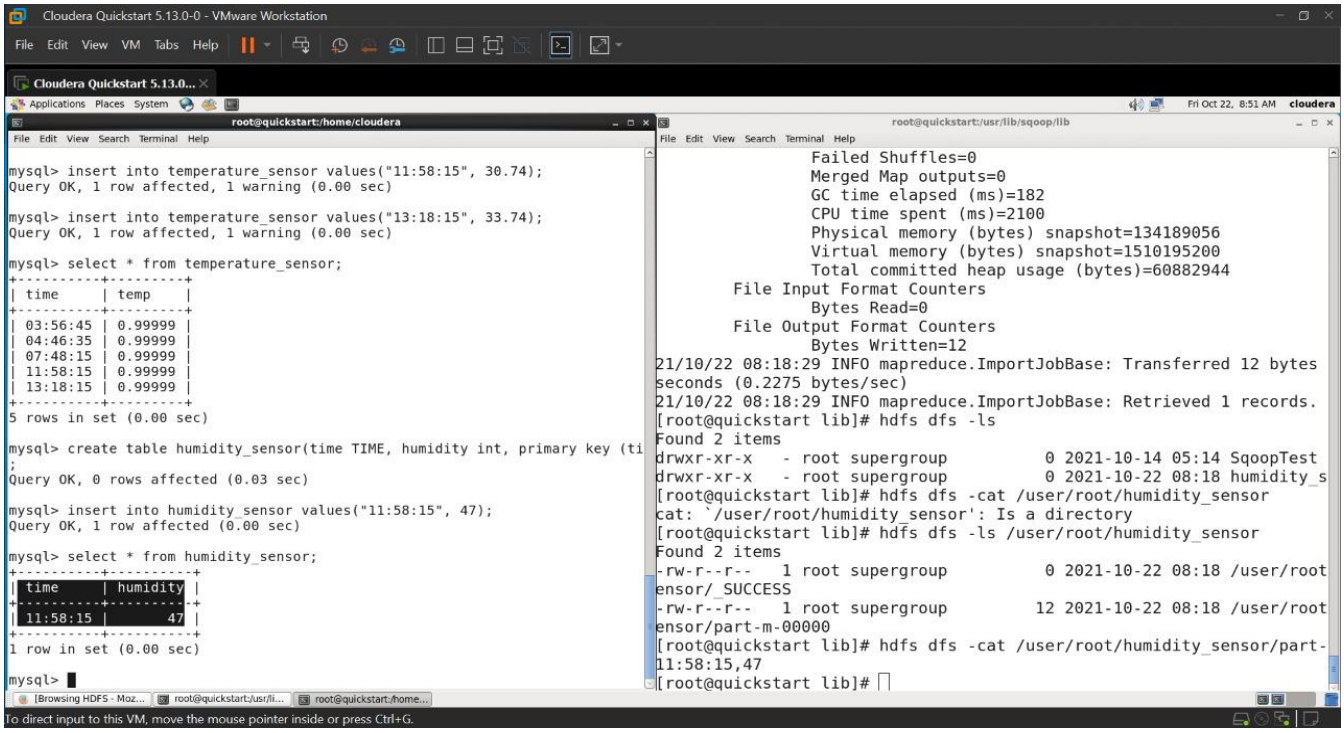
MySQL

Tableau (Data Visualization)

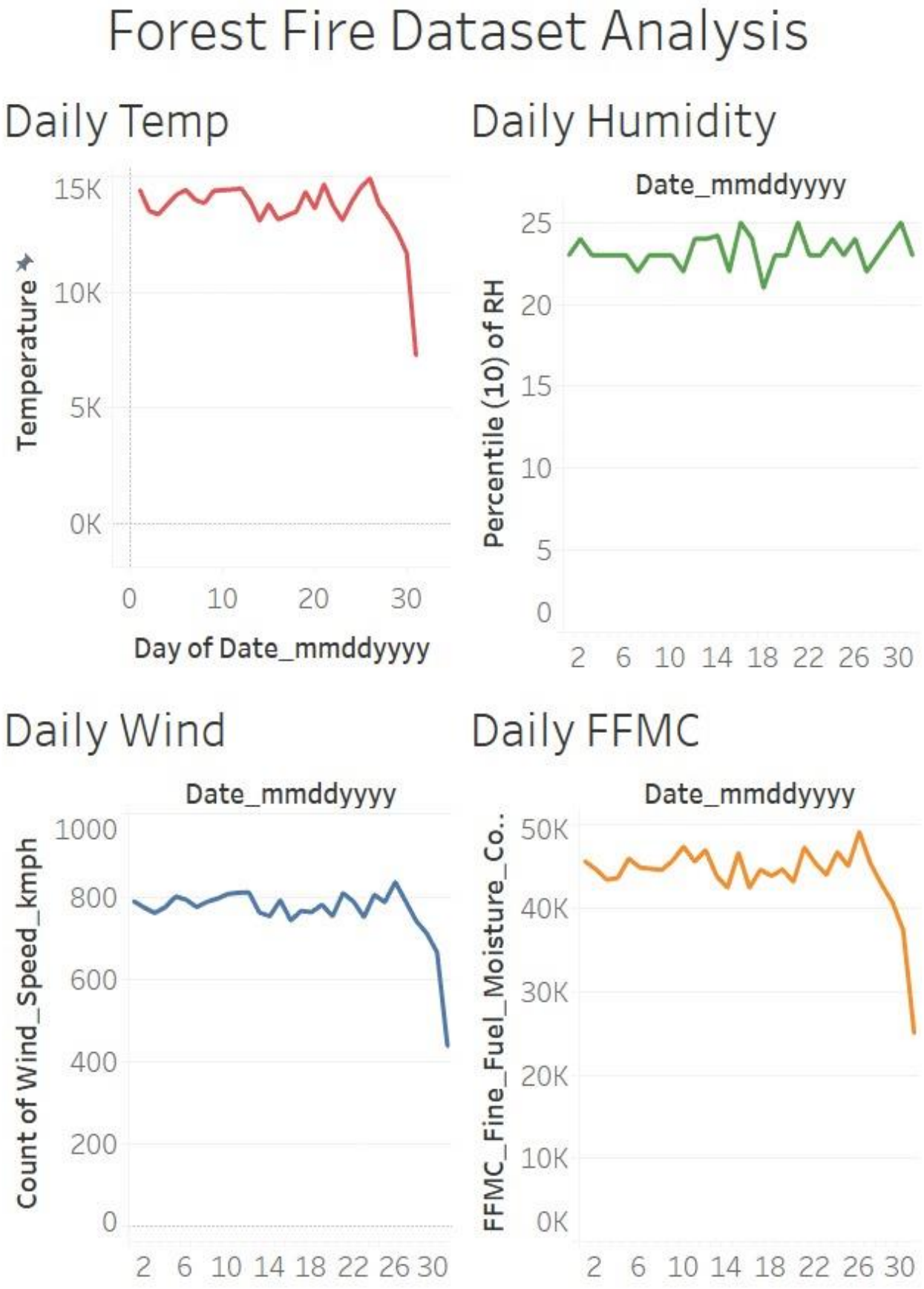
Pydoop

# RESULTS

## 1. Data Migration from censors to HDFS using Apache Sqoop



## 2. Data Visualization using Tableau



# OBSERVATIONS

According to some performance parameters we found that the processing of MapRed works fine. We also have other options to carry out further in the project like processing with Apache Hive. By changing some hyperparameters we managed to get precision as high as possible.

# CONCLUSION

This concept is just a basic one. The further development of the project can be done in a larger configuration. We can use GSM module along with Nano-satellite / Pico-satellite (small satellite) which is very small compared to normal satellite. A group/cluster of Pico satellites may be used to receive the data from the transmitter which is at the forest region and then transmitting to the ground station. When we are using cluster of small satellites, we may also consider using small satellites for different purposes like monitoring the forest fire, vegetation of the forest, for monitoring the climate changes of that particular area like this in low cost. Since the proposed project is easy for the installation due to the simple arrangements, the time frame for developing and integrating the sub-systems and installation process is less. Even for the demonstration purpose we can put up the fire manually in any plain land and we can test for the working of these components.

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